Experiment No:09

Python for Data 25: Chi-Squared Tests

back to index

Last lesson we introduced the framework of statistical hypothesis testing and the t-test for investigating differences between numeric variables. In this lesson, we turn our attention to a common statistical test for categorical variables: the chi-squared test.

Chi-Squared Goodness-Of-Fit Test

In our study of t-tests, we introduced the one-way t-test to check whether a sample mean differs from the an expected (population) mean. The chi-squared goodness-of-fit test is an analog of the one-way t-test for categorical variables: it tests whether the distribution of sample categorical data matches an expected distribution. For example, you could use a chi-squared goodness-of-fit test to check whether the race demographics of members at your church or school match that of the entire U.S. population or whether the computer browser preferences of your friends match those of Internet uses as a whole.

When working with categorical data, the values themselves aren't of much use for statistical testing because categories like "male", "female," and "other" have no mathematical meaning. Tests dealing with categorical variables are based on variable counts instead of the actual value of the variables themselves.

Let's generate some fake demographic data for U.S. and Minnesota and walk through the chi-square goodness of fit test to check whether they are different:

```
import numpy as np
import pandas as pd
import scipy.stats as stats
```

```
national = pd.DataFrame(["white"]*100000 + ["hispanic"]*60000 +\
                        ["black"]*50000 + ["asian"]*15000 + ["other"]*35000)


minnesota = pd.DataFrame(["white"]*600 + ["hispanic"]*300 + \
                         ["black"]*250 +["asian"]*75 + ["other"]*150)

national_table = pd.crosstab(index=national[0], columns="count")
minnesota_table = pd.crosstab(index=minnesota[0], columns="count")
print( "National")
print(national_table)
print(" ")
```

```
print( "Minnesota")
print(minnesota_table)
```

```
National
col_0      count
0
asian      15000
black      50000
hispanic   60000
other      35000
white     100000

Minnesota
col_0     count
0
asian        75
black       250
hispanic    300
other       150
white       600
```

Chi-squared tests are based on the so-called chi-squared statistic.

You calculate the chi-squared statistic with the following formula:

sum((observed−expected)2expected)

In the formula, observed is the actual observed count for each category and expected is the expected count based on the distribution of the population for the corresponding category. Let's calculate the chi-squared statistic for our data to illustrate:

```
observed = minnesota_table

national_ratios = national_table/len(national)  # Get population ratios

expected = national_ratios * len(minnesota)   # Get expected counts

chi_squared_stat = (((observed-expected)**2)/expected).sum()

print(chi_squared_stat)
```

```
col_0
count    18.194805
dtype: float64
```

```
crit = stats.chi2.ppf(q = 0.95, # Find the critical value for 95% confidence
                      df = 4)   # Df = number of variable categories - 1

print("Critical value")
print(crit)

p_value = 1 - stats.chi2.cdf(x=chi_squared_stat,  # Find the p-value
                             df=4)
```

```
print("P value")
print(p_value)
```

```
Critical value
9.487729036781154
P value
[0.00113047]
```

```
stats.chisquare(f_obs= observed,   # Array of observed counts
                f_exp= expected)
```

```
Power_divergenceResult(statistic=array([18.19480519]),
pvalue=array([0.00113047]))
```

Chi-Squared Test of Independence

Independence) is a key concept in probability that describes a situation where knowing the value of one variable tells you nothing about the value of another. For instance, the month you were born probably doesn't tell you anything about which web browser you use, so we'd expect birth month and browser preference to be independent. On the other hand, your month of birth might be related to whether you excelled at sports in school, so month of birth and sports performance might not be independent.

The chi-squared test of independence tests whether two categorical variables are independent. The test of independence is commonly used to determine whether variables like education, political views and other preferences vary based on demographic factors like gender, race and religion. Let's generate some fake voter polling data and perform a test of independence:

```
np.random.seed(10)

# Sample data randomly at fixed probabilities
voter_race = np.random.choice(a= ["asian","black","hispanic","other","white"
                              p = [0.05, 0.15 ,0.25, 0.05, 0.5],
                              size=1000)

# Sample data randomly at fixed probabilities
voter_party = np.random.choice(a= ["democrat","independent","republican"],
                              p = [0.4, 0.2, 0.4],
                              size=1000)

voters = pd.DataFrame({"race":voter_race,
                      "party":voter_party})

voter_tab = pd.crosstab(voters.race, voters.party, margins = True)

voter_tab.columns = ["democrat","independent","republican","row_totals"]

voter_tab.index = ["asian","black","hispanic","other","white","col_totals"]
```

```
observed = voter_tab.iloc[0:5,0:3]   # Get table without totals for later us
voter_tab
```

	democrat	independent	republican	row_totals
asian	21	7	32	60
black	65	25	64	154
hispanic	107	50	94	251
other	15	8	15	38
white	189	96	212	497
col_totals	397	186	417	1000

```
expected =  np.outer(voter_tab["row_totals"][0:5],
                     voter_tab.loc["col_totals"][0:3]) / 1000

expected = pd.DataFrame(expected)

expected.columns = ["democrat","independent","republican"]
expected.index = ["asian","black","hispanic","other","white"]

expected
```

	democrat	independent	republican
asian	23.820	11.160	25.020
black	61.138	28.644	64.218
hispanic	99.647	46.686	104.667
other	15.086	7.068	15.846
white	197.309	92.442	207.249

```
chi_squared_stat = (((observed-expected)**2)/expected).sum().sum()

print(chi_squared_stat)
```

```
7.169321280162059
```

```
crit = stats.chi2.ppf(q = 0.95, # Find the critical value for 95% confidence
                      df = 8)   # *

print("Critical value")
print(crit)

p_value = 1 - stats.chi2.cdf(x=chi_squared_stat,  # Find the p-value
                             df=8)
```

```
print("P value")
print(p_value)
```

```
Critical value
15.50731305586545
P value
0.518479392948842
```

```
stats.chi2_contingency(observed= observed)
```

```
Chi2ContingencyResult(statistic=np.float64(7.169321280162059),
pvalue=np.float64(0.518479392948842), dof=8, expected_freq=array([[ 23.82 ,
11.16 ,  25.02 ],
       [ 61.138,  28.644,  64.218],
       [ 99.647,  46.686, 104.667],
       [ 15.086,   7.068,  15.846],
       [197.309,  92.442, 207.249]]))
```

The output shows the chi-square statistic, the p-value and the degrees of freedom followed by the expected counts.

As expected, given the high p-value, the test result does not detect a significant relationship between the variables.

1.When do we go for Chi-square test?

The Chi-squared test is primarily used for categorical data. There are two main types:

Chi-squared Goodness-of-Fit Test: Used to determine if the observed distribution of a single categorical variable in a sample matches an expected or theoretical distribution (e.g., does the distribution of colors in a bag of candies match what the manufacturer claims?). Chi-squared Test of Independence: Used to determine if there is a statistically significant association between two categorical variables (e.g., is there a relationship between gender and preferred type of car?). **2.Do we use Chi-square test for independence? Explain How?**

Yes, the Chi-squared test of independence is specifically designed for this purpose. It works by comparing the observed frequencies in a contingency table (a table that displays the frequency distribution of the variables) with the expected frequencies. The expected frequencies are what you would anticipate if there were no relationship (i.e., if the variables were independent).

How it works: You calculate a Chi-squared statistic based on the differences between observed and expected counts. A large difference indicates that the observed data deviates significantly from what would be expected under independence, suggesting a relationship between the variables. **3.Explain role of Critical value and p-value in Chi-square test.**

Both the critical value and p-value are used to make a decision about your hypothesis:

Critical Value: This is a threshold value obtained from the Chi-squared distribution table for a given significance level (alpha, commonly 0.05) and degrees of freedom. If your calculated Chi-squared statistic is greater than the critical value, you reject the null hypothesis (which states that there is no relationship/difference). P-value: The p-value is the probability of observing a test statistic as extreme as, or more extreme than, the one calculated from your sample, assuming the null hypothesis is true. If the p-value is less than your chosen significance level (alpha, e.g., 0.05), you reject the null hypothesis. A small p-value indicates strong evidence against the null hypothesis. In essence, both methods lead to the same conclusion: if your test statistic is in the **4.Can we use the Chi-squared test to show the existence of correlation between 2 features in a dataset? How will you interpret test results to indicate correlation presence? **

Yes, the Chi-squared test of independence is used to assess if there's an association (or dependence) between two categorical features. While 'correlation' often implies a linear relationship between numeric variables, in the context of categorical variables, an association or dependence is the equivalent concept. If two categorical variables are associated, it means that the distribution of one variable differs depending on the value of the other variable.

Interpreting test results for correlation/association: When you perform a Chi-squared test of independence, your null hypothesis ($H_0 H_0$) is that there is no association between the two categorical variables (i.e., they are independent). The alternative hypothesis ($H_1 H_1$) is that there is an association between them. If the p-value is less than your chosen significance level (e.g., 0.05), you reject the null hypothesis. This suggests that there is statistically significant evidence to conclude that an association exists between the two categorical features in your dataset. In simpler terms, the features are likely related. If the Chi-squared statistic is greater than the critical value, you also reject the null hypothesis, leading to the same conclusion.
5.What are degrees of freedom and how to calculate it while performing a Chi-squared test?

Degrees of Freedom (df) refers to the number of independent pieces of information that go into calculating a statistic. In simpler terms, it's the number of values in a study that are free to vary. It's a crucial component because it helps determine the shape of the Chi-squared distribution, which in turn affects the critical value and p-value.

Calculation for Chi-squared Goodness-of-Fit Test: For a goodness-of-fit test, where you are comparing observed frequencies of a single categorical variable to expected frequencies: df = (number of categories - 1) For example, if you have 5 categories

(like 'asian', 'black', 'hispanic', 'other', 'white' in our earlier example), the degrees of freedom would be 5 - 1 = 4.

Calculation for Chi-squared Test of Independence: For a test of independence, where you are examining the relationship between two categorical variables in a contingency table: df = (number of rows - 1) * (number of columns - 1) Here, 'rows' and 'columns' refer to the number of categories for each of the two categorical variables being tested (excluding any total rows or columns). For instance, if you have a 5x3 contingency table (like our voter race vs. party example with 5 races and 3 parties), the degrees of freedom would be (5 - 1) * (3 - 1) = 4 * 2 = 8.

Double-click (or enter) to edit